

Homework rules

- Group discussion is *highly recommended*. But you have to write down your solution by *yourself*. Aim for clarify and conciseness in your solution, emphasizing the main ideas over low-level details.
- You should *cite* all the references that help you to solve the homework, including the people you have discussed with and the online material you looked at.
- You are required to type your solution using latex and submit a *pdf* to Gradescope and a *tex* file to NTUCool. You may type your solution in English (preferable) or Chinese.
- You are not expected to solve *all* problems. Just try your best! If you don't solve a problem, try your best to write down your thinking process.
- You may use AI however you wish to deepen your understanding of the lecture material. Upload the notes, talk to your AI about them, ask for more explanation or examples; it's all fine. You may not use LLMs in any way to work on your homework. You may not upload assignments, ask for hints, ask how certain concepts from the lectures might be applied to specific homework problems, or upload your assignments to check for correctness or clarity or anything else. You may not include any AI generated content whatsoever in your homework submissions (This policy is taken from MIT 6.5620).
- We will use a *random-checking policy* in which students may be asked to present their homework solutions during TA office hours, with the expectation that everyone will present *at least once*. Evaluation is based on *consistency*, meaning students must be able to reproduce the solutions they originally submitted. If a student fails this consistency check, they will receive a significant *percentage deduction* from their homework score.
- 作業總分為22分，可得最大計分為22分。
- Deadline 6/14(日)23:59，遲交時間最晚6/16(二) 23:59 (遲交成績打9折)，超過則以0 分計算。

1. (10 points) **(Facts about SIS)**

- (a) (3 points) **(Short integer solution exists.)** Let $q(n)$ be a prime, $m > n \log q$, $\beta > \sqrt{m}$, show that for all $A \in \mathbb{Z}_q^{n \times m}$, there exists some short vector $r \in \mathbb{Z}^m$, such that $Ar = 0 \pmod q$, $\|r\|_2 \leq \beta$, and $r \neq 0^m$.
- (b) (3 points) **(Inhomogeneous SIS is harder than SIS.)** The *Inhomogeneous Short Integer Solution (ISIS)* problem $\text{ISIS}_{n,m,q,\beta}$ with parameters n, m, q, β is the problem that given $A \leftarrow \mathbb{Z}_q^{n \times m}$ and $y \leftarrow \mathbb{Z}_q^n$, find a “short” vector $r \in \mathbb{Z}^m$ such that $Ar = y \pmod q$ and $\|r\|_2 \leq \beta$.

As with the SIS problem, for sufficiently large q , $m > n \log q$, and $\beta > \sqrt{m}$, ISIS is believed to be hard, namely, for all PPT algorithms Adv there is a negligible function $\mu(\cdot)$ such that for every $n \in \mathbb{N}$,

$$\Pr[A \leftarrow \mathbb{Z}_q^{n \times m}, y \leftarrow \mathbb{Z}_q^n : r \in \mathbb{Z}^m \leftarrow \text{Adv}(1^n, A, y) \text{ such that } (Ar = y \pmod q) \wedge (\|r\|_2 \leq \beta)] \leq \mu(n).$$

Show that, there exists some constant c such that for $\hat{m} > m$ and $\hat{\beta} = \beta + c$ such that $\text{SIS}_{n,\hat{m},q,\hat{\beta}}$ is hard implies $\text{ISIS}_{n,m,q,\beta}$ is hard. In other words, please show that if there is an efficient solver for the $\text{ISIS}_{n,m,q,\beta}$ problem then there is an efficient algorithm that solves the $\text{SIS}_{n,\hat{m},q,\hat{\beta}}$ problem.

- (c) (4 points) **(SIS is harder than DLWE.)** Let $q = n^{10}$, $\alpha = n^{-5}$, $m > n \log q$, $\beta = O(\sqrt{n \log q})$ and χ be the discrete Gaussian distribution with deviation $\alpha \cdot q$. Show that if $\text{DLWE}_{n,m,q,\chi}$ is hard, then $\text{SIS}_{n,m,q,\beta}$ is hard. In other word, show that if there is an efficient solver for $\text{SIS}_{n,m,q,\beta}$ problem then there is an efficient algorithm that solves the $\text{DLWE}_{n,m,q,\chi}$ problem.

2. (6 points) **(HVZK with abort is sufficient for Fiat-Shamir Signature)**

In class, we introduced a lattice-based identification scheme which only satisfies a much weaker security notion that allows the prover to abort, and the aborted transcript is not required to be simulatable. While it may seem odd to aim for such weak notion, this is in fact sufficient to construct a signature scheme from such ID scheme.

Let us first formally formulate the ID scheme with abort. An ID scheme with abort consist of the following four algorithms $(\text{KeyGen}, P_1, P_2, V)$. For $(pk, sk) \leftarrow \text{KeyGen}(1^n)$, the prover with input (pk, sk) and the verifier with input pk interacts as follow:

- (Prover Phase): the prover P_1 takes the input (pk, sk) and output the commitment com and a private state st .
- (Verifier Phase): the verifier V samples a challenge $\text{ch} \leftarrow \mathcal{C}$. Here for simplicity we only consider $\mathcal{C}$ with *super-polynomial* size, i.e. $\frac{1}{|\mathcal{C}|} = \text{negl}(n)$
- (Prover Phase): the prover P_2 takes the input $(pk, sk, \text{st}, \text{ch})$ and output resp to the verifier V . Here resp may be the abort symbol $\perp$.
- (Verifier Phase): If $\text{resp} = \perp$, the verifier V aborts with output $\perp$. Otherwise, the verifier V verify the response resp with $V(pk, \text{com}, \text{ch}, \text{resp}) \rightarrow 0/1$, which is a deterministic algorithm that outputs 1 if it accepts and 0 otherwise. The verifier returns $V(pk, \text{com}, \text{ch}, \text{resp})$.

An ID-scheme with abort should satisfy the following two properties:

- **Completeness with abort.** There exist some abort rate $p(n) \leq 1 - \frac{1}{\text{poly}(n)}$ and some negligible function $\mu(n)$, such that for all $n \in \mathbb{N}$,

$$\Pr[(pk, sk) \leftarrow \text{KeyGen} : \langle P(pk, sk), V(pk) \rangle = 0] \leq \mu(n)$$

and

$$\Pr[(pk, sk) \leftarrow \text{KeyGen} : \langle P(pk, sk), V(pk) \rangle = \perp] \leq p(n)$$

- **Security with aborting oracle.** Consider the following security experiment $\text{ID}^{\text{forge}, \perp}(n)$:
 - $(pk, sk) \leftarrow \text{KeyGen}(1^n)$, send pk to Adv
 - $\text{com} \leftarrow \text{Adv}^{\text{Trans}_{sk}^{\perp}(\cdot)}(pk)$
 - $\text{ch} \leftarrow \mathcal{C}$, send ch to Adv
 - $\text{resp} \leftarrow \text{Adv}^{\text{Trans}_{sk}^{\perp}(\cdot)}(pk, \text{ch})$
 - Adv wins if $V(pk, \text{com}, \text{ch}, \text{resp}) = 1$.

Where the $\text{Trans}_{sk}^{\perp}(\cdot)$ oracle is an oracle that, on called, runs a honest interaction of $\langle P(pk, sk), V(pk) \rangle$ and obtains a transcript $T = (\text{com}, \text{ch}, \text{resp})$. If $\text{resp} \neq \perp$, $\text{Trans}_{sk}^{\perp}(\cdot)$ returns T . Otherwise, it returns a single $\perp$ (and does not return (com, ch)).

An ID scheme is secure with abort if for all probabilistic polynomial-time adversaries Adv , there exists a negligible function $\mu(\cdot)$ such that: for each $n \in \mathbb{N}$,

$$\Pr[\text{Adv wins in } \text{ID}^{\text{forge}, \perp}(n)] \leq \mu(n).$$

Finally, to achieve the security of a signature, we need a small extra property from our underlying ID scheme:

- **High first-message min-entropy.** Given $(pk, sk) \leftarrow \text{KeyGen}(1^n)$, we say that $P(pk, sk)$ has first-message min-entropy $H_{P_1, \infty}(pk, sk) = r$ if for all possible commitment $\widetilde{\text{com}} \leftarrow P_1(pk, sk)$,

$$\Pr[(\text{com}, \text{st}) \leftarrow P_1(pk, sk) : \text{com} = \widetilde{\text{com}}] \leq 2^{-r}$$

We say that an ID-scheme has high first-message min-entropy if there exist some r such that $2^{-r} = \text{negl}(n)$, and that

$$\Pr[(pk, sk) \leftarrow \text{KeyGen}(1^n) : H_{P_1, \infty}(pk, sk) < r] \leq \text{negl}(n)$$

Now, given a ID-scheme with abort, We can obtain a signature scheme under the random oracle model via the Fiat-Shamir transformation. The *Fiat-Shamir signature scheme* $\text{Sig} = \text{FS}[\text{ID}_\perp; H] = (\text{KeyGen}, \text{Sign}, \text{Vrfy})$ derived from the ID-scheme with abort $\text{ID}_\perp = (\text{KeyGen}, P_1, P_2, V)$ works as follows:

- **KeyGen:** the same as KeyGen from ID scheme;
- To sign a message m using a secret key sk , the signing algorithm **Sign** runs as:
 - let $p(n)$ be the abort rate of the underlying ID-scheme. Set $t = n \times \frac{1}{1-p(n)}$
 - Run t following trails:
 - * Run prover $P_1(pk, sk)$, obtaining a commitment com and a state st ;
 - * computes the challenge $ch \leftarrow H(m, \text{com})$, where H is the random oracle;
 - * Run $P_2(pk, sk, st, ch)$, obtaining a response resp .
 - * If $\text{resp} \neq \perp$, Output signature $\sigma = (\text{com}, \text{resp})$. Otherwise, give up the current trial and retry.
 - If all t trails fail, output $\perp$
- the verification algorithm **Vrfy**: to verify a signature $\sigma = (\text{com}, \text{resp})$ on a message m using a pk , it computes $ch \leftarrow H(m, \text{com})$ and output $V(pk, \text{com}, ch, \text{resp})$.

Show that, if $\text{ID}_\perp$ is a secure ID-scheme with abort that additionally has high first-message min-entropy, then $\text{Sig} = \text{FS}[\text{ID}_\perp; H]$ is a secure signature scheme under the random oracle model.

(**Hint.** The proof should be *very similar* to the standard Fiat-Shamir proof taught in class. However, you might need to take care with a few extra details.)

3. **(6 points, Lamport Signature is secure)** In this problem, we will show that we can in fact construct signatures solely from one-way functions. Consider the security definition of 1-time signature: The one-time signature experiment $\text{Sig} - \text{forge}_{A, \Pi}^{1\text{-time}}(n)$

- $\text{KeyGen}(1^n)$ is run to obtain keys (pk, sk) .
- Adversary A is given pk and asks a single query m' to its oracle $\text{Sign}_{sk}(\cdot)$. A then outputs (m, σ) .
- The output of the experiment is defined to be 1 if and only if $\text{Vrfy}_{pk}(m, \sigma) = 1$ and $m \neq m'$.

Signature scheme $\Pi = (\text{KeyGen}, \text{Sign}, \text{Vrfy})$ is existentially unforgeable under a single-message attack, or is a one-time signature scheme, if for all probabilistic polynomial-time adversaries A , there exists a negligible function $\mu(\cdot)$ such that:

$$\Pr[\text{Sig} - \text{forge}_{A, \Pi}^{1\text{-time}}(n) = 1] \leq \mu(n).$$

The Lamport's Signature is described as following: Let $H : \{0, 1\}^n \rightarrow \{0, 1\}^n$ be a OWF. Construct a signature scheme for messages of length $l = l(n)$ as follows:

- **KeyGen:** on input 1^n , proceed as follows for $i \in \{1, \dots, l\}$: Choose uniform $x_{i,0}, x_{i,1} \in \{0, 1\}^n$. Compute $y_{i,0} := H(x_{i,0})$ and $y_{i,1} := H(x_{i,1})$. Output the public key $pk = \{y_{i,0}, y_{i,1}\}_{i=1}^l$ and the private key $sk = \{x_{i,0}, x_{i,1}\}_{i=1}^l$.

- Sign: on input a private key sk as above and a message $m \in \{0, 1\}^l$ with $m = m_1 \cdots m_l$, output the signature $(x_{1,m_1}, \dots, x_{l,m_l})$.
- Vrfy: on input a public key pk as above, a message $m \in \{0, 1\}^l$ with $m = m_1 \cdots m_l$, and a signature $\sigma = (x_1, \dots, x_l)$, output 1 if and only if $H(x_i) = y_{i,m_i}$ for all $1 \leq i \leq l$.

Show that the above construction is a one-time signature scheme.